

Ecogo Project Proposal for B2C Flight & Hotel Mobile App

Prepared for



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COMPANY PROFILE

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Ecogo Software Solutions Pvt Ltd is an emerging technology company offering end-to-end IT solutions, product development, and strategic consulting in the fields of Travel Technology, Artificial Intelligence, Internet of Things (IoT), and Enterprise Software Development.

We specialize in building scalable B2B and B2C travel booking platforms, seamless API integrations with global providers, and AI-driven digital products. Our vision of “Innovation with Reliability” drives us to deliver cost-effective, high-performance solutions that empower businesses to scale globally.

INTRODUCTION

Ecogo Software Solutions Pvt Ltd is pleased to present this proposal to Digital Wings for the development of a modern B2C mobile application for Flights and Hotels. This solution is designed to empower Digital Wings with a user-centric digital platform that enhances customer experience and supports business growth.

The objective of this project is to deliver a seamless travel booking experience across flights and hotels through a mobile-app digital solution. The solution will provide Digital Wings' customers with smooth booking flows, intuitive navigation, and reliable booking services.

CONFIDENTIALITY STATEMENT

This document is the property of Ecogo Software Solutions Pvt Ltd and is intended solely for the evaluation by Digital Wings. It contains commercially sensitive information and must not be shared, copied, or distributed outside the parties without prior written consent from Ecogo Software Solutions Pvt Ltd and Digital Wings. Unauthorized disclosure or use of this document may result in legal consequences and disciplinary actions.

PROPOSED SOLUTION

The proposed solution is to design and implement a **B2C mobile application** for **Digital Wings** covering the following modules:

- **Flights:** Mobile Application (UI/UX + Frontend + Backend Development with API Integration)
- **Hotels:** Mobile Application (UI/UX + Frontend + Backend Development with API Integration)

FUNCTIONAL REQUIREMENTS

Module Features Table

MODULE	SUBMODULE	SPECIFIC FEATURE	REMARKS
Static to every module			
Home page	About Us	Home Page and Inner Page, Banners Implementation	
	Management	Top Menu Implementation with submenu support	Menu Items will be flights and hotels (all these pages are explained below)
		About Us	
		Terms & conditions	
		Privacy Policy	PDF managed from CMS will be shown here
		Fare Rules	PDF managed from CMS will be shown here

	Footer	User Agreement	PDF uploaded from backend will be shown to users
		Help & FAQs	PDF uploaded from backend will be shown to users
		Contact Us	Quick Contact Info
		Testimonials	Menu layout and position
		News	Quick News Module
User Registration & Login	Create an Account	User Registration and Profile Set Up: - Mobile Number OTP on mobile number for verification	
	Login	User Validation and Secured Login - Login id (Mobile Number) - OTP	
	Social Login	Integration of Facebook, Google+, Apple	
	My Account	Dashboard showing all trips details with upcoming, completed and cancelled Trip.	Clicking on a particular trip will allow users to see details of the trip. Also, users can

User Account		Includes My Bookings, View Cancellations, My Profile, Travelers Support	see the complete details of the trip such as in out time, reason of cancellation etc. This is a repository of all the bookings done via app such as flights, hotels.
	My Profile	Details of Travelers Uploaded documents, Passport information, Profile settings, Edit Profile	
	Manage Bookings	View, Print, Email Tickets/Cancel Bookings, Change Travel Date, Check Refund Status	
		Year, from date, to date, PNR, Trip id, service id Share: Using this option, the user will be able to share the trip details (hotel, flight, experience with booking) with their friends and family via	

		message, social media platforms	
		Make a Payment, Submit Account Details	
	Check Refund Status	User can view refund status for his cancel/upgrade trips	
Flight Module			
Search	Search Dashboard	Search by One or more parameters (Type of Trip, Location, Date)	
	Advanced Search	Departure Time, Return Time, Airline Preference	
	Filter Search	User can apply filter with one or more parameters(No of Stops, Time frame, Airlines, Price Range	
	Modify Search	User can make a modification in the search parameters such as type of trip (one way, multicity,	

		round trip), destination, origin, no. of travellers, departs on, arrives on, class(economy, business),	
	Calendar Option	User can choose single or multiple date for his trip.	
	Fare Calendar	Cheapest Flights for the month will be shown to user for trip plan	
	Fare Comparison Table	User can view flight fares with calendar option	
Utility	Display all airport information	Departure Airport, Return Airport, Connecting airports, Appy filters on airports	
	Display cabin facilities Display flight and gate information	In flight meal, Wi-Fi filters Layover timing, arrival timing, depart timing, gates information	Based API/Data availability

	Dynamic Currency Conversation		
Booking Process	Detail Price and tax summary within booking process / purchase path	User can check the price break down and tax for his trip booking	Fare rule and Fare Pricing is included where user will be shown breakup of all the taxes, base fare, discounts applicable on the flight booking
	Seat Plan	Users can check the seat plan of the flights while booking	
	Create multiple passenger booking		
Customer Experience	Special Service Request (SSR)	Special Meal and paid meal menu options offered. Stored for bookings made on the App or for logged-in users, so user can easily manage booking such as updating seats and meals.	

	Add Favorites	Allow user to save booking and other functionalities/pages to a personal favorites list that can be stored on the device	
	Calendar	To include a Timetable Facility that allows users to view all scheduled flight times	
Payment	Process Credit Card and Debit card payments		Payment gateway
	Process mobile forms of payment The adoption of universal mobile payment systems		
Localization	Multilanguage		Each Language plugin Based language and
	currency		currency based on APIs
Notification	Notifications:	Remind traveller to check-in if they have	

		not done so already	
	Notifications: Baggage Policy	Give travellers info about cabin and carry-on baggage allowances for their flight	
	Notification: Realtime alerts	Travel alerts	
	Notification: Rich messaging	Deep links – place users right into specific screen or flows / from mobile app	
Deal/Discount	Offer Code	facility, which also allows for time sensitive offers facilitating promotions	
Hotels			
Hotel Booking Management	Search	The user will be able to search the Hotels using this option. The user will have to enter information like location along with the check-in check out details followed by No of rooms, no of adults	

		and no of children. The user will be able to save the search they have done and that will be also shown in recent searches.	
	Modify Search	The user will be able to modify search on the basis of Price, Star Rating, Hotel name, location, Rate Type etc.	
	Hotel Details	As soon as the user will click on any hotel in the list view, a detailed description will be displayed to the user like hotel overview, reviews, amenities, check in check out time and policy information.	
	Integration with Map API	The application will be integrated with Google/Other Map provider Maps. The user will be able to view the location of Hotel on Maps	Users can book via map as they will be able to see a pinpoint of all hotels on map and on enlarged view of these locations, user can see book now option with form to

			book.
	Review/Rating	Using this option, the user will be able to view the reviews of people and ratings given by them to the Hotel	
	Room Selection	* The user will be able to select the Hotel from a list of Hotels. Once the user will select the Hotel he/she will get the option for Room selection where he will be able to view the various plans and associated information. * Plan Selection * Booking Policy	
	List View/Map View	The user will be able to view the Hotels in List View and Map View with Image Gallery	
	Passenger Information	The user will feed in the passenger information like first name, last name,	

		mobile no and e-mail for making booking	
	Review Booking	Once the user will feed all the information, he/she will be able to view the booking form for cross verifying the details or changing any detail.	
	Enter Coupon Code	As the customer will make selection and proceed for booking, he/ she can enter the coupon code is they have to avail discounts.	
	Select Payment	Once the user will be done with flight/hotel he/she will be asked for the payment mode like CC/DB/ net banking and etc and user will be able to make purchase.	Based on payment gateway

Travel Stories		This section will have the stories from people who have visited any place. The users will share their experience, rating etc on a platform which would be viewable by the other users as well. The users will also be able to create their stories.	
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NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the **attributes and quality standards** of the application, such as **security, scalability, and usability**. These also include constraints or restrictions on the design and development process.

Some characteristics of non-functional requirements are:

- They define the overall qualities or attributes of the resulting system.
- They place restrictions on the product being designed and the process used to build it.
- They specify external constraints the product must meet.
- Examples include security, usability, performance, and compatibility.
- Project management aspects (cost, time, and schedule) are often linked to non-functional requirements but are handled separately.

The following non-functional requirements are applicable for the proposed **Digital Wings Flights & Hotels Mobile Application**:

Compatibility

- The mobile application will be developed for **Android and iOS platforms**, ensuring compatibility with the **latest two OS versions**.
- The backend will be accessible through **REST APIs**, ensuring easy integration with **TBO APIs** and future third-party services.

Security

- The application will have strong authentication and authorization mechanisms to ensure only verified users can access services.
- All communication will be encrypted using **HTTPS/TLS**.
- Protection against malicious attacks (e.g., SQL Injection, Cross-Site Scripting, CSRF) will be implemented following **OWASP Top 10 guidelines**.

Scalability

- The mobile application and backend services will be designed using a **modular and scalable architecture**.
- Microservice-ready architecture ensures future additions (e.g., Holidays, Cars, Visa) can be integrated without affecting Flights & Hotels.
- Auto-scaling on the backend will be supported to handle peak traffic efficiently.

Nonfunctional requirements describe the quality and attributes of the system. These ensure that the application is secure, scalable, user-friendly, and reliable.

Performance

The mobile application must be **interactive, responsive, and optimized** to minimize delays in every user action. The following requirements are considered under performance:

- Short response time for any action or request.
- High throughput and **high availability** of the application and backend services.
- Efficient **caching mechanisms** to reduce repeated load times.

- Optimized mobile design ensuring **quick load time** and smooth navigation.
- Targeted content delivery for improved user experience.
- **Mobile responsiveness** and adaptability across devices.

Assumptions

1. Ecogo will share **UI/UX design** for the mobile application, and development will proceed based on client feedback and approval.
2. Content beyond English (e.g., multi-language requirements) will be provided by **Digital Wings**.
3. Costs arising from any **third-party integrations or service providers** (SMS, payment gateway, etc.) will be handled separately and billed additionally.
4. Any external **software licenses, plugins, or hardware** required will be provided by **Digital Wings**.
5. **Data categorization and handling guidelines** will be shared by Digital Wings to ensure proper storage and labelling.
6. **Flights and Hotels APIs will be provided by TBO**, which Ecogo will integrate into the platform. (Other modules like Holidays, Visa, Forex, Cars are **out of current scope**).
7. Ecogo will develop the application **from scratch**, including design, UI/UX, and frontend + backend development.
8. All **design assets (logos, branding, PSDs)** will be provided by Digital Wings.
9. Ecogo will provide up to **one design iterations** during the UI/UX stage.

Out Of Scope

- Development of new APIs (only **TBO APIs will be integrated**).
- Content standardization across languages.
- Migration of legacy backend systems.
- Migration of old content/data.

- Non-flight/hotel modules (Visa, Forex, Cars, Holidays, Bus, etc. — may be considered in later phases).

DEVELOPMENT DETAILS

Component	Technology / Tools	Remarks
Mobile Application	Flutter (Dart) / React Native	Cross-platform (Android & iOS) mobile-first development
Backend APIs	Go Lang	REST APIs with integration to TBO APIs (Flights & Hotels)
Database	PostgreSQL / MySQL	Structured storage for bookings, users, payments
Caching	Redis	For session management, search caching, performance optimization
Authentication	JWT / OAuth2	Secure login, token-based access
Payment Gateway	Razor pay / Stripe / PayPal	PCI-DSS compliant multi-currency payments
Hosting / Infra	AWS / Kubernetes	Scalable, high-availability cloud deployment
Notifications	Firebase Cloud Messaging	Push notifications for bookings & offers
3rd Party Integration	TBO API (Flights & Hotels)	Real-time availability, pricing, booking

PROJECT TIMELINE

Phase	Duration	Deliverables
Requirement Gathering & Design	2 weeks	Wireframes, user journeys, AI interaction flows, high-fidelity prototypes
Backend Development & LLM Integration	2 weeks	Profile analysis, AI rules, itinerary manager, admin dashboard
Mobile App Development (Flutter)	4 weeks	iOS & Android app with conversational AI & itinerary manager
Testing & Quality Assurance	1 weeks	Functional testing, AI personalization validation, compliance & security testing
Deployment & Launch	1 week	Release to App Store & Play Store
Total Estimated Duration	Phase 1: Android version to be delivered within 2 months from project commencement. Phase 2: iOS version to be delivered within 3 months from project commencement.	

PROJECT COST & COMMERCIALS

Based on the tentative project plan, the proposed cost is as follows:

S. No.	Description	Price (USD)
1.	Design & Development of B2C Mobile Application covering: - Flights: Mobile App (UI/UX + Frontend + Backend + TBO API Integration) - Hotels: Mobile App (UI/UX + Frontend + Backend + TBO API Integration) Includes UI/UX design, API integration, booking flows, payments, and notifications.	USD 12000.00

TERMS & CONDITIONS

1. After detailed requirement gathering, the **effort and project timeline may vary** (typical variance 15–20%).
2. Any **scope changes or change requests** will be estimated and billed separately with prior approval from **Digital Wings**.
3. Any additional modules beyond **Flights & Hotels (TBO integration)** will be considered out of scope and billed separately.
4. Additional API supplier integrations will be charged in the range of USD 2,500.00 to USD 3,500.00 per integration.

Ecogo Software Solutions Pvt Ltd will provide **technical support** for the mentioned project under the **Standard Support Contract** as follows:

- **Till 3 month after UAT completion:** Technical support will be **free of cost** if delivered from Ecogo's development center. Any required **onsite support** will incur additional cost.
- Ecogo Software Solutions Pvt Ltd's Standard Terms & Conditions shall apply.
- **Quarterly / Annual Maintenance Contract (AMC):**
A Maintenance Contract (AMC) may be executed upon completion of the warranty period. The pricing for the AMC will be calculated at 15% of the total project invoice value.
- Flight booking includes integration with one provider. Additional providers will be charged separately.
- Hotel booking includes integration with one hotel API provider. Any additional provider integrations beyond this scope will be charged separately.
- Any additional scope of work or support beyond the agreed deliverables will be billed separately, as per standard charges. Additional support and development hours will be charged at USD 35.00 per hour.

PAYMENT TERMS

- **40%** Upon Project Confirmation (Kickoff)
- **30%** Upon UI/UX Design Approval
- **20%** Upon Completion of Frontend Development

- 5% Upon Completion of Backend & API Development
- 5% Upon Final Deployment & Project Handover

PROJECT MANAGEMENT STRATEGY

Discovery & Planning Phase (Inception)

The purpose of the inception phase is to identify the **functional, non-functional, and performance requirements**, define the **look and feel** of the mobile application, finalize **navigation flows**, and prepare a **comprehensive requirements document** for the project.

During this phase, Ecogo will deploy a **dedicated Business Analyst, UX/UI Designer, and Technical Lead** (as required) to work with the stakeholders of **Digital Wings** and capture requirements.

All requirements will be standardized into **use cases** with main flows, alternate flows, exceptions, pre-conditions, and post-conditions. In addition, **user stories** will be written to ensure that the key scenarios for **Flights and Hotels booking with TBO integration** are fully considered.

Each requirement will be analyzed for:

- **Completeness** – all requirements are captured, documented, and signed off.
- **Testability** – requirements are unambiguous and measurable through test cases.
- **Usability** – functionality is user-friendly and intuitive.
- **Validation** – user input checks and business rules are clearly defined.
- **Performance & Scalability** – requirements for load, response times, concurrent users, and system expansion are validated.

The **Technical Lead** will assess the **feasibility** of each requirement and provide implementation notes for complex features (e.g., seat maps, TBO API flows). The estimated effort will be compared against expected timelines to ensure successful delivery.

End Deliverables of this Phase:

- Scope of Work / Requirements Document
- Creative Brief (branding, visual design references)
- User Interaction Diagrams
- High-Level Architectural Design
- Creative Assets
- Detailed Design Document
- Signed-off Project Plan

Conceptualization & Design

The purpose of this phase is to create **prototypes (mock-ups/wireframes)** for the **Flights & Hotels mobile application** and to complete the **high-level architectural design** for the system. The **Project Manager, Technical Lead, and UI/UX Designer** will be involved in this phase as per project needs.

A **modular approach** will be taken in the design process, allowing development to begin on approved components while design work continues on remaining modules.

In addition to architecture, the solution design will include creation and sharing of **interactive prototypes** for approval by Digital Wings. The **Technical Lead** will verify that the architecture supports all documented **functional and non-functional requirements**.

The architecture will be reviewed for:

- **Modularity** – ensuring components are independent and reusable.
- **Portability** – compatibility with multiple mobile platforms (Android, iOS).
- **Extensibility** – ability to integrate new modules in future (e.g., Holidays, Cars).
- **Security & Performance** – identifying vulnerabilities or bottlenecks early, with remedial measures incorporated.

Where necessary, **Proof of Concepts (POCs)** will be constructed to validate assumptions (e.g., seat maps, TBO API workflows).

The detailed architecture will define services, components, and their interfaces. The database model will be mapped to these components, and strategies for exception handling and logging will be documented for each service.

Deliverables for this Phase:

- High-Level Architectural Design
- Interactive UI/UX Prototypes & Wireframes
- Creative Assets (icons, branding elements)
- Completed Detailed Design Document

Development Phase

The purpose of this phase is to **build the components of the mobile application** as per the planned sprints. The **Technical Lead, Mobile Developers, Backend Developers, and QA Engineers** will be actively involved. All developers and testers working on this project will follow **Ecogo's best practices in coding, version control, testing, and security compliance**.

Coding Structure

- Design documents will be finalized before any code is written.
- Standardized coding guidelines will be followed, ensuring consistent quality.
- Checklists will be used for all parts of code and database design.
- Code reviews will be conducted by senior engineers to maintain robustness.

Version Control

- All source code, design documents, and configuration files will be maintained in **Git-based version control** from day one.
- Branching, tagging, and merging will follow industry-standard GitFlow practices.

- Secure repositories (GitHub/Bitbucket/GitLab) will be used with restricted access.

Bug Reporting and Resolution

- Bug tracking will be done via **JIRA or equant system** (not through emails).
- All bugs, enhancements, and change requests will be tracked in the issue management system.
- Each issue will include detailed **replication steps and severity classification**.

Security Model

- The application will run entirely on **HTTPS** after login, ensuring secure transactions.
- Sensitive data (e.g., payment details) will not be logged in plain text.
- Passwords and tokens will be stored using **bcrypt or equivalent encryption standards**.

Testing Phase

This phase begins with the preparation of a **test plan**, followed by execution once development is complete. The QA team ensures that the application meets **business objectives, functional accuracy, and technical requirements**.

Testing will cover:

- Functional testing of flight & hotel booking flows.
- Performance and device compatibility testing (Android/iOS).
- Integration testing with **TBO APIs**.
- Security and compliance checks

Deliverables: Test Plan, Executed Test Cases, Quality Report, and Staged Deployment for UAT.

Roles & Responsibilities

- **QA Manager** – Oversees quality process, release & delivery.

- **QA Lead** – Defines scope, prepares test plan, manages RTM.
- **QA Engineers** – Create & execute test cases, log defects, verify fixes.

User Acceptance Testing (UAT)

Ecogo will involve **Digital Wings** in UAT to validate the system against real business scenarios before final deployment.

ROLES & RESPONSIBILITIES

Roles for both Ecogo and Digital Wings are defined as below:

Role	Responsibility
Product Owner (Digital Wings)	Business inputs, approvals
Scrum Master (Ecogo PM)	Agile process, sprint planning
Tech Lead (Ecogo)	Backend architecture & development
UX Lead (Ecogo)	UI/UX design
QA Lead (Ecogo)	Testing & UAT support

CONCLUSION

Ecogo Software Solutions Pvt Ltd would like to conclude this business proposal with the following highlights

- Ecogo has provided a unique and comprehensive solution for the development of the B2C Flights & Hotels Mobile Application.
- If implemented, the solution will enable Digital Wings to achieve its business objectives in a cost-efficient manner.

- The greatest success for Ecogo lies in the satisfaction and confidence of our clients. We are open to reviewing and discussing proposal elements such as scope, timeline, and team structure to ensure Digital Wings' comfort.
- This engagement will foster a strong and long-term partnership between Digital Wings and Ecogo, driving innovation and growth in the travel technology ecosystem.

